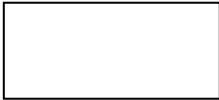
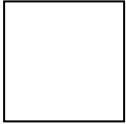
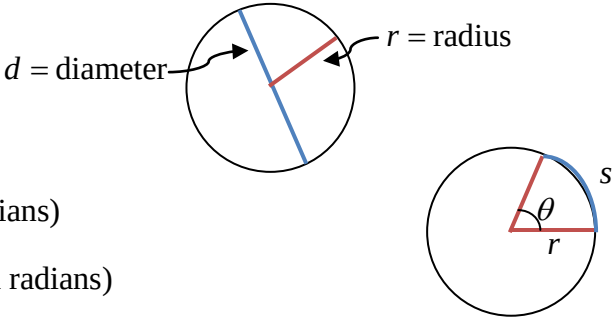
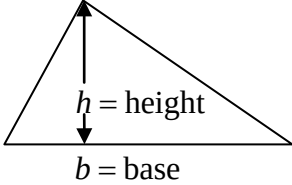
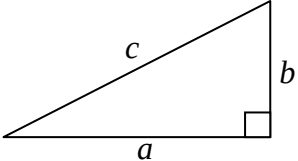
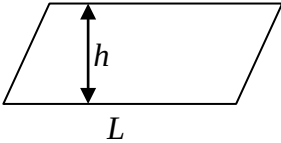
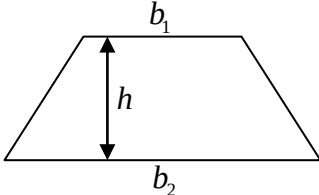
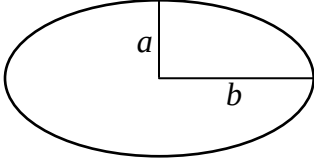
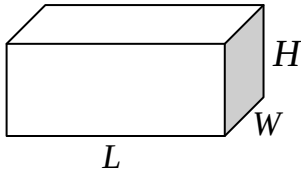
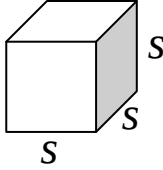
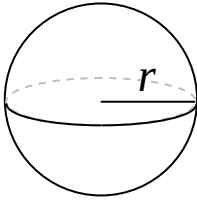
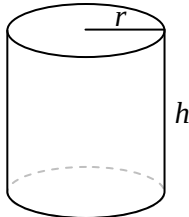
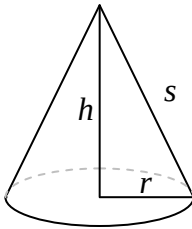
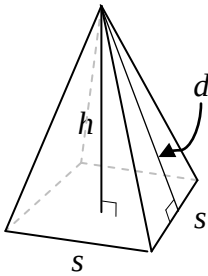
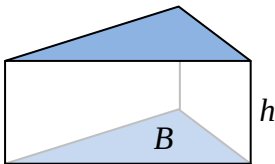
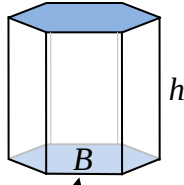


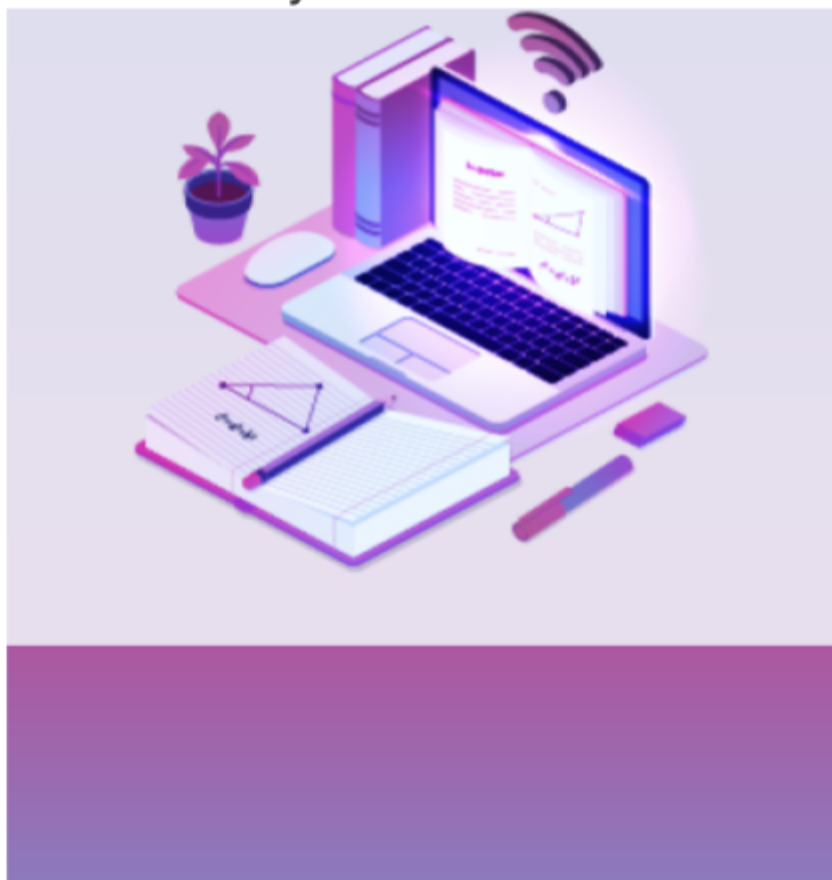
Geometric Formulas

Shape	General Formulas	Picture
Rectangles	Perimeter: $P = 2L + 2W$ Area: $A = L \cdot W$	 $W = \text{width}$ $L = \text{length}$
Squares	Perimeter: $P = 4S$ Area: $A = S \cdot S = S^2$	 $S = \text{side}$
Circles	Circumference: $C = 2\pi r = \pi d$ Area: $A = \pi r^2 = \frac{\pi}{4} d^2$ Arc Length: $s = r \cdot \theta$ (θ in radians) Area of Sector: $A_s = \frac{1}{2} r^2 \cdot \theta$ (θ in radians)	
Triangles	Area: $A = \frac{1}{2} b \cdot h$	 $h = \text{height}$ $b = \text{base}$
Right Triangles	Side Length Relationship: $c^2 = a^2 + b^2$	
Parallelograms	Area: $A = L \cdot h$	
Trapezoids	Area: $A = \frac{1}{2} (b_1 + b_2) \cdot h$	
Ellipses	Area: $A = \pi \cdot a \cdot b$	

Shape	General Formulas	Picture
Rectangular Solids (Boxes)	Surface Area (with top and bottom): $SA = 2LW + 2LH + 2WH$ Volume: $V = L \cdot W \cdot H$	
Cubes	Surface Area (with top and bottom): $SA = 6s^2$ Volume: $V = s \cdot s \cdot s = s^3$	
Spheres	Surface Area: $SA = 4\pi r^2$ Volume: $V = \frac{4}{3}\pi r^3$	
Cylinders	Surface Area (with top and bottom): $SA = 2\pi rh + 2\pi r^2$ Volume: $V = \pi r^2 h$	
Cones	Surface Area (with bottom): $SA = \pi rs + \pi r^2$ Volume: $V = \frac{1}{3}\pi r^2 h$	
Pyramids (Square-based)	Surface Area (with bottom): $SA = 2ds + s^2$ Volume: $V = \frac{1}{3}s^2 h$	
Prisms	Surface Area (with top and bottom): $SA = 2B + h \cdot (\text{Perimeter of base})$ Volume: $V = B \cdot h$	  $B = \text{Area of Base}$

GMAT Geometry Formulas PDF

By GMATPoint.com



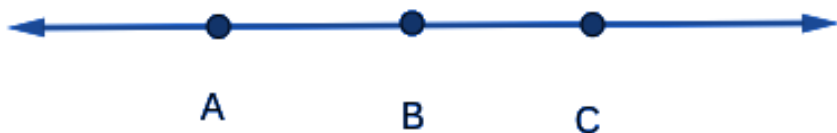
GMAT Geometry Formulas

- Geometry is often considered to be one of the hardest sections to crack without preparation; however, that is not necessarily the case - with consistent preparation, geometry questions would become easier to handle.
- With so many formulas to learn and remember, this section is going to take a lot of time to master.
- A good way to utilize this formula collection: read a formula, try to visualize it and solve as many questions related to the formula as you can.
- Remember that knowing a formula and knowing when or where to apply it are two different abilities. You could gain the first ability by going through reading this formulae list, but the latter will come only through solving problems.
- In this document, we provide an exhaustive list of GMAT Geometry formulas to make the geometry section a lot easier for GMAT aspirants.

Lines and Angles

Collinear points:

Three or more points lying on the single straight line. In this diagram the three points A, B and C are collinear



Concurrent lines:

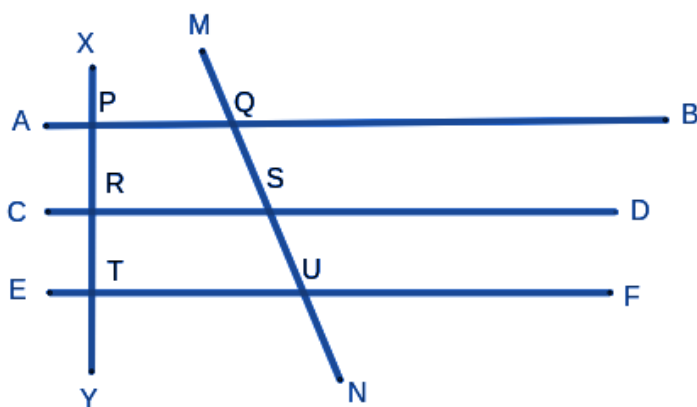
If three or more lines lying in the same plane intersect at a single point then those lines are called concurrent lines.

The three lines X, Y and Z are concurrent lines here.

- When two angles A and B are complementary,
sum of A and B is 90°
- When two angles A and B are supplementary,

sum of A and B is 180°

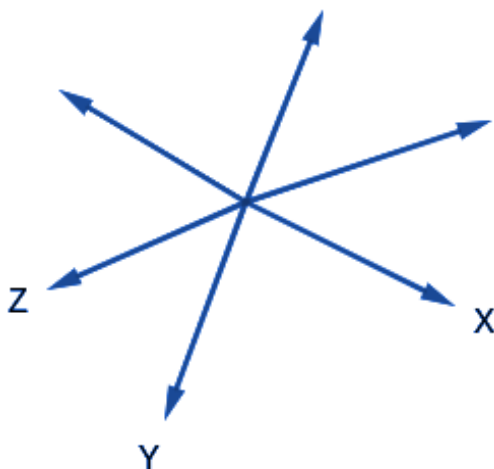
- When two lines intersect, opposite angles are equal. Adjacent angles are supplementary
- When any number of lines intersect at a point, the sum of all the angles formed = 360°
- Consider parallel lines AB, CD and EF as shown in the figure.



- XY and MN are known as transversals
- $\angle XPQ = \angle PRS = \angle RTU$ as corresponding angles are equal

- Interior angles on the side of the transversal are supplementary. i.e. $\angle PQS + \angle QSR = 180^\circ$
- Exterior angles on the same side of the transversal are supplementary. i.e. $\angle MQB + \angle DSU = 180^\circ$
- Two transversals are cut by three parallel lines in the same ratio i.e. $\frac{PR}{RT} = \frac{QS}{SU}$

Co-ordinate Geometry



- The distance between two points with coordinates (X_1, Y_1) , (X_2, Y_2) is given by

$$D = \sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2}$$

- Slope, $m = \frac{y_2 - y_1}{x_2 - x_1}$ (If $x_2 = x_1$ then the lines are perpendicular to each other)

- Mid point between two points $A(x_1, y_1)$ and $B(x_2, y_2)$

is $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

- When two lines are parallel, their slopes are equal

i.e. $m_1 = m_2$

- When two lines are perpendicular, product of their

slopes = -1 i.e, $m_1 * m_2 = -1$

- If two intersecting lines have slopes m_1 and m_2 then the angle between two lines will be

$$\tan \theta = \frac{m_1 - m_2}{m_1 m_2}$$

(where θ is the angle between the lines)

- The length of perpendicular from a point (X_1, Y_1) on

the line $AX+BY+C = 0$ is
$$P = \frac{AX_1 + BY_2 + C}{\sqrt{A^2 + B^2}}$$

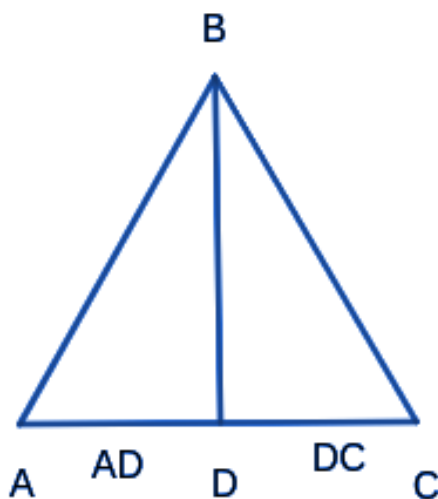
- **Equations of a lines :**

General equation of a line	$Ax + By = C$
Slope intercept form	$y = mx + c$ (<i>c is y intercept</i>)
Point-slope form	$y - y_1 = m(x - x_1)$
Intercept form	$\frac{x}{a} + \frac{y}{b} = 1$
Two point form	$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$

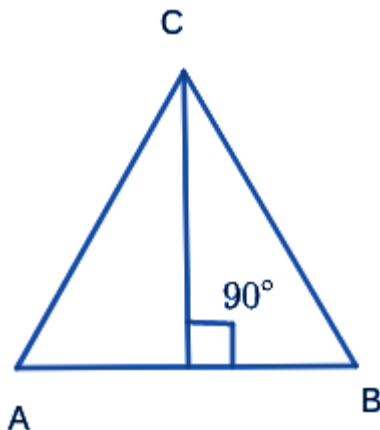
Triangles

- Sum of all angles in a triangle is 180°
- An angle less than 90° is called an acute angle.
An angle greater than 90° is called an obtuse angle.
- A triangle with all sides unequal is called scalene triangle
- A triangle with two sides equal is called an isosceles triangle.
- The two angles of an isosceles triangle that are not contained between the equal sides are equal
- A triangle with all sides equal is called an equilateral triangle. All angles of an equilateral triangle equal 60° .
- If in a triangle all of its angles are less than 90° than that triangle is called as acute angled triangle
- A triangle with one of its angle equal to 90° than that triangle is called as Right angled triangle

- A triangle with one of its angles greater than 90° than that triangle is called an Obtuse angled triangle.
- If one side of a triangle is produced then that exterior angle formed is equal to the sum of opposite remote interior angles
- A line joining the mid point of a side with the opposite vertex is called a median. (Here D is the midpoint of the AC side or $AD = DC$). BD is the median of this triangle ABC.



- A perpendicular drawn from a vertex to the opposite side is called the altitude



- A line that bisects and also makes right angle with the same side of the triangle is called perpendicular bisector
- A line that divides the angle at one of the vertices into two parts is called angular bisector
- All points on an angular bisector are equidistant from both arms of the angle.
- All points on a perpendicular bisector of a line are equidistant from both ends of the line.

- In an equilateral triangle, the perpendicular bisector, median, angle bisector and altitude (drawn from a vertex to a side) coincide.
- The point of intersection of the three altitudes is the Orthocentre.
- The point of intersection of the three medians is the centroid.
- The three perpendicular bisectors of a triangle meet at a point called the Circumcentre. A circle drawn from this point with the circumradius would pass through all the vertices of the triangle.
- The three angle bisectors of a triangle meet at a point called the incentre of a triangle. The incentre is equidistant from the three sides and a circle drawn from this point with the inradius would touch all the sides of the triangle.
- Sum of any two sides of a triangle is always greater than its third side

- Difference of any two sides of a triangle is always lesser than it's third side
- **Pythagoras theorem:**

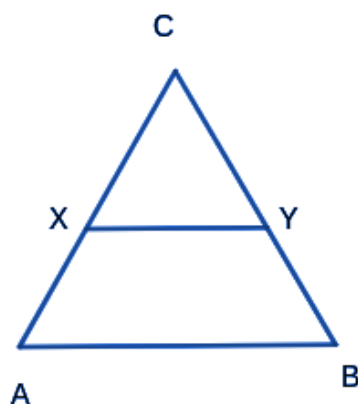
In a right angled triangle ABC where

$$\angle B = 90^\circ, AC^2 = AB^2 + BC^2$$

- **Mid Point Theorem :**

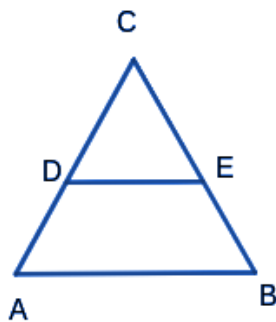
The line joining the midpoint of any two sides in a triangle is parallel to the third side and is half the length of the third side. If X is the midpoint of CA and Y is the midpoint of CB.

Then XY will be parallel to AB and $XY = \frac{1}{2} * AB$



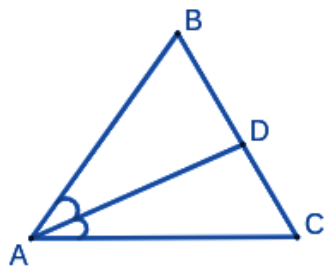
- **Basic proportionality theorem :**

If a line is drawn parallel to one side of a triangle and it intersects the other two sides at two distinct points then it divides the two sides in the ratio of respective sides. If in a triangle ABC, D and E are the points lying on AB and BC respectively and DE is parallel to AC then $AD/DB = EC/BE$



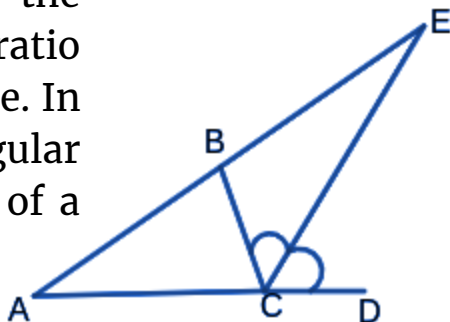
- **Interior Angular Bisector theorem :**

In a triangle the angular bisector of an angle divides the side opposite to the angle, in the ratio of the remaining two sides. In a triangle ABC if AD is the angle bisector of angle A then AD divides the side BC in the same ratio as the other two sides of the triangle.
i.e. $BD/CD = AB/AC$.



● Exterior Angular Bisector theorem :

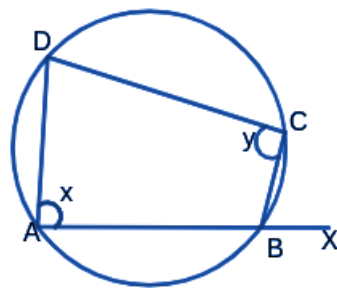
The angular bisector of the exterior the angle of a triangle divides the opposite side externally in the ratio of the sides containing the angle. In a triangle ABC, if CE is the angular bisector of exterior angle BCD of a triangle, then $AE/BE = AC/BC$



● Cyclic Quadrilateral :

If a quadrilateral has all its vertices on the circle and its opposite angles are supplementary (here $x+y = 180^\circ$) then that quadrilateral is called cyclic quadrilateral. In a cyclic quadrilateral the opposite angles are supplementary.

Exterior angle is equal to its remote interior opposite angle. (here $\angle CBX = \angle ADC$)



- If x is the side of an equilateral triangle then the

$$\text{Altitude (h)} = \frac{\sqrt{3}}{2} x$$

$$\text{Area} = \frac{\sqrt{3}}{4} x^2$$

$$\text{Inradius} = \frac{1}{3} * h$$

$$\text{Circumradius} = \frac{2}{3} * h$$

Similar triangles :

If two triangles are similar then their corresponding angles are equal and the corresponding sides will be in proportion.

For any two similar triangles :

- Ratio of sides = Ratio of medians = Ratio of heights =
Ratio of circumradii = Ratio of Angular bisectors
- Ratio of areas = Ratio of the square of the sides.
Tests of similarity : (AA / SSS / SAS)

Congruent triangles:

If two triangles are congruent then their corresponding angles and their corresponding sides are equal.

Tests of congruence : (SSS / SAS / AAS / ASA)

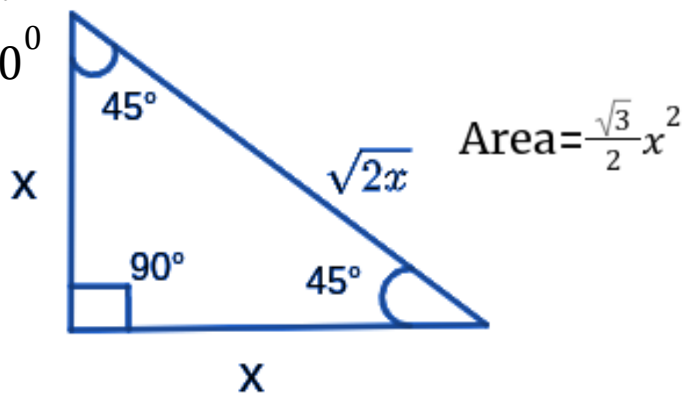
Area of a triangle:

- $A = \sqrt{s(s - a)(s - b)(s - c)}$ where $s = \frac{(a+b+c)}{2}$
- $A = \frac{1}{2} * \text{base} * \text{altitude}$
- $A = \frac{1}{2} * ab * \sin C$ (C is the angle formed between sides a and b)
- $A = \frac{abc}{4R}$ where R is the circumradius

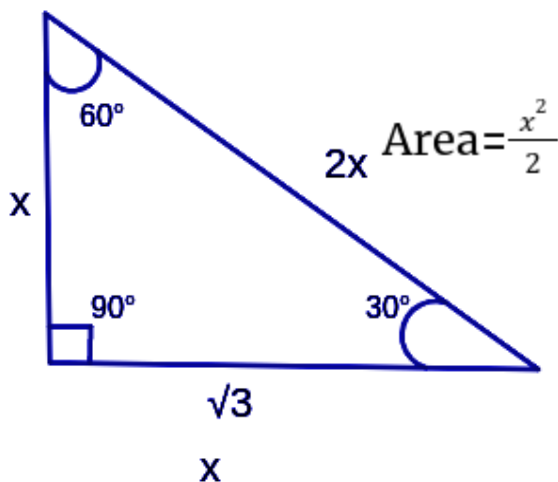
- $A = r * s$ where r is the inradius and s is the semi perimeter. (where a , b and c are the lengths of the sides BC , AC and AB)

Special triangles :

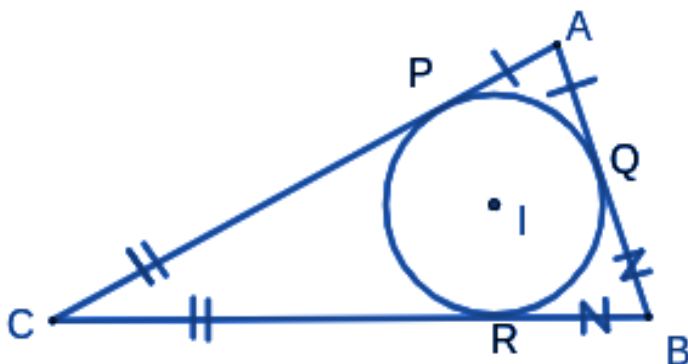
- $30^\circ, 60^\circ, 90^\circ$



- $45^\circ, 45^\circ, 90^\circ$



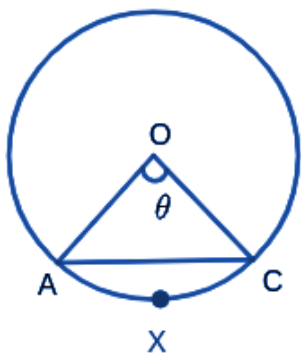
- Consider the triangle ABC with incentre I, and the incircle touching the triangle at P, Q, R as shown in the diagram. As tangents drawn from a point are equal, $AP=AQ$, $CP=CR$ and $BQ=BR$.



- In an equilateral triangle, the centroid divides the median in the ratio 2:1. As the median is also the perpendicular bisector, angle bisector, G is also the circumcentre and incentre.
- If a is the side of an equilateral triangle, circumradius $= a/\sqrt{3}$ and inradius $= a/(2\sqrt{3})$

Circles

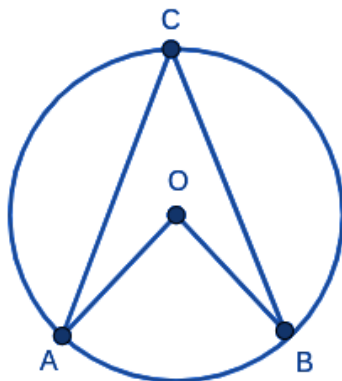
- The angle subtended by a diameter of circle on the circle = 90^0
- Angles subtended by an equal chord are equal. Also, angles subtended in the major segment are half the angle formed by the chord at the center
- Equal chords of a circle or equidistant from the center
- The radius from the center to the point where a tangent touches a circle is perpendicular to the tangent
- Tangents drawn from the same point to a circle are equal in length
- A perpendicular drawn from the center to any chord, bisects the chord



$$\text{Area of sector OAXC} = \frac{\theta}{360} * \pi r^2$$

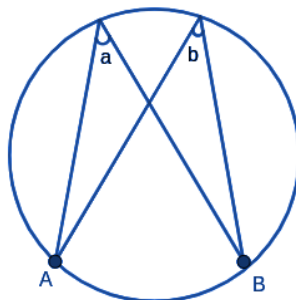
$$\text{Area of minor segment AXC} = \frac{\theta}{360\pi r^2} - \frac{1}{2r^2 \sin \theta}$$

Inscribed angle Theorem :



$$2 \angle ACB = \angle AOB$$

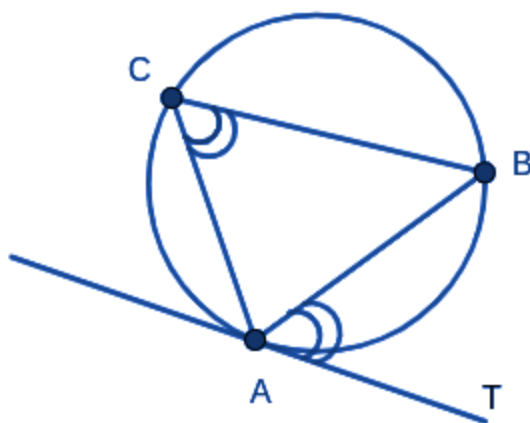
The angle inscribed by the two points lying on the circle, at the center of the circle, is twice the angle inscribed at any point on the circle by the same points.



Angles subtended by the same segment on the circle will be equal. So, here angles a and b will be equal.

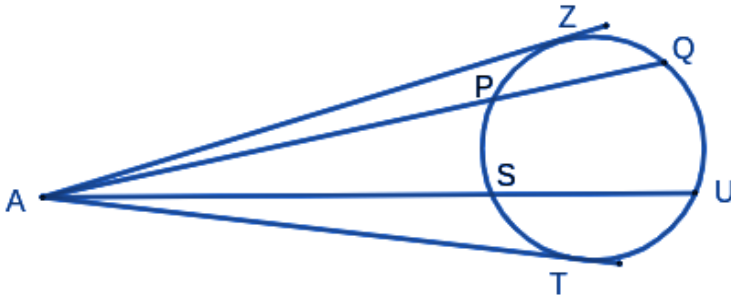
The angle made by a chord with a tangent to one of the ends of the chord is equal to the angle subtended by the chord in the other segment.

As shown in the figure, $\angle ACB = \angle BAT$.



Consider a circle as shown in the image.

Here, $AP * AQ = AS * AU = AT^2$

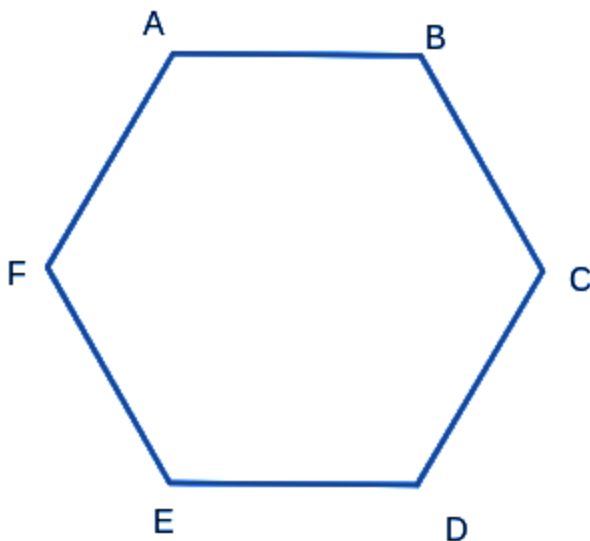


Two tangents drawn to a circle from an external common point will be equal in length. So here $AZ = AT$

Polygons and Quadrilaterals

- If all sides and all angles are equal, then the polygon is a regular polygon
- A regular polygon of n sides has $\frac{n(n-3)}{2}$ diagonals
- In a regular polygon of n sides, each exterior angle is $\frac{360}{n}$ degrees.
- Sum of measure of all the interior angles of a regular polygon is $180 (n-2)$ degrees (where n is the number of sides of the polygon)

- Sum of measure of all the exterior angles of regular polygon is 360 degrees



ABCDEF is a regular hexagon with each side equal to 'x' then

- Each interior angle = 120°
- Each exterior angle = 60°
- Sum of all the exterior angles = 360°
- Sum of all the interior angles = 720°
- Area = $\frac{3\sqrt{3}}{2}a^2$

Areas of different geometrical figures:

Triangles	$\frac{1}{2} * base * height$
Rectangle	$length * width$
Trapezoid	$\frac{1}{2} * sum\ of\ bases * height$
Parallelogram	$base * height$
Circle	$\pi * radius^2$
Rhombus	$\frac{1}{2} * product\ of\ diagonals$
Square	$side^2\ or\ \frac{1}{2}diagonals^2$
Kite	$\frac{1}{2} * product\ of\ diagonals$

Solids

Volume of different solids:

Cube	$length^3$
Cuboid	$length * base * height$
Prism	$Area\ of\ base * height$
Cylinder	$\pi r^2 h$
Pyramid	$\frac{1}{3} * Area\ of\ base * height$
Cone	$\frac{1}{3} * \pi r^2 * h$
Cone Frustum (If R is the base radius, r is the upper surface radius and h is the height of the frustum)	$\frac{1}{3} * \pi(R^2 + Rr + r^2)$
Sphere	$\frac{4}{3} * \pi * r^3$
Hemi-sphere	$\frac{2}{3}\pi r^3$

Total Surface area of different solids:

Prism	$2 * \text{base area} * \text{base perimeter} * \text{height}$
Cube	$6 * \text{length}^2$
Cuboid	$2(lh + bh + lb)$
Cylinder	$2\pi rh + 2\pi r^2$
Pyramid	$\frac{1}{2} * \text{Perimeter of base} * \text{slant height} + \text{Area of base}$
Cone (l is the slant height)	$\pi r(l + r)$
Cone Frustum (where R & r are the radii of the base faces and l is the slant height)	$\pi r(R^2 + r^2 + Rl + rl)$
Sphere	$4\pi r^2$
Hemi-sphere	$3\pi r^2$

Lateral/Curved surface area:

Prism	$\text{base perimeter} * \text{height}$
Cube	$4 * \text{length}^2$
Cuboid	$2 \text{ length} * \text{height} + 2 \text{ breadth} * \text{height}$
Cylinder	$2\pi rh$
Pyramid	$\frac{1}{2} * \text{Perimeter of base} * \text{slant height}$
Cone (l is the slant height)	πrl
Cone Frustum (where R is the base radius, l is the slant height)	$\pi(R + r)L$



Telegram



[GMATPoint Telegram channel](#)

Subscribe To [GMATPOINT YouTube Channel](#)



[Free GMAT Preparation Online](#)



Take free GMAT Daily Targets With Video Solutions:
<https://gmatpoint.com/gmat-daily-target>

Functions and Rules:

If a and r are real numbers and $r \neq 0$, then

$$\sum_{j=0}^n ar^j = \begin{cases} \frac{ar^{n+1} - a}{r - 1} & \text{if } r \neq 1 \\ (n+1)a & \text{if } r = 1. \end{cases}$$

$$\sum_{k=1}^n k$$

$$\frac{n(n+1)}{2}$$

$$\sum_{k=1}^n k^2$$

$$\frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=1}^n k^3$$

$$\frac{n^2(n+1)^2}{4}$$

$$\sum_{k=0}^{\infty} x^k, |x| < 1$$

$$\frac{1}{1-x}$$

$$\sum_{k=1}^{\infty} kx^{k-1}, |x| < 1$$

$$\frac{1}{(1-x)^2}$$

- **LCM * GCD** = the multiple of the 2 numbers

- Sum of odd numbers : $\left(\frac{n + n \% 2}{2}\right)^2$

- Sum of even numbers: $\frac{(\frac{n}{2} * (2 + n - n \% 2))}{2}$

- Quadratic Formula : $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Area

Area square = S^2

Area rectangle = $L * W$

Area triangle = $\frac{1}{2} b h$

$$= \sqrt{s(s-a)(s-b)(s-c)} \quad s = \text{half perimeter}$$

$$= \frac{1}{2} a b \sin(c)$$

$$= \frac{ax(by-cy)+bx(cy-ay)+cx(ay-by)}{2}$$

Area parallelogram = $b h$ **Area trapezoid** = $\frac{1}{2} h (b_1 + b_2)$

Area rhombus = $\frac{1}{2} d_1 * d_2$

Area circle = πr^2

Circumference = $2 \pi r$

Area sphere = $4 \pi r^2$

Area regular polygon = $\frac{1}{2} a * p$ a is the radius, p is perimeter

Area of irregular polygon =

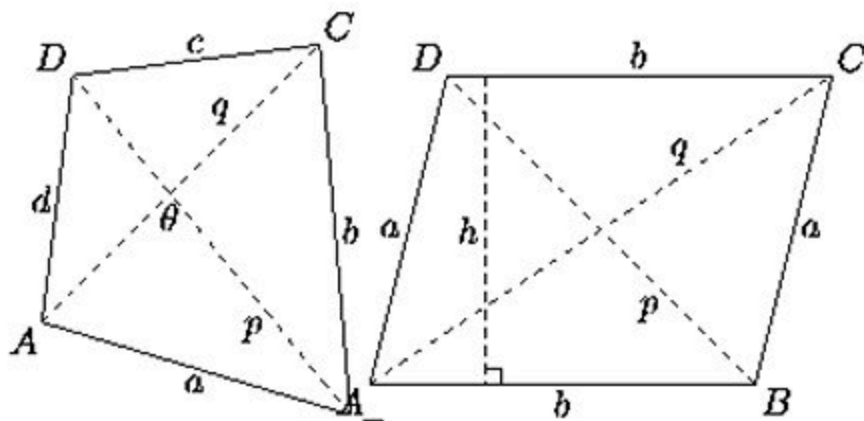
List of x, y coordinates in counter clockwise

$$= \frac{(x_1 y_2 - y_1 x_2) - (x_2 y_3 - y_2 x_3) \dots + (x_n y_1 - y_n x_1)}{2}$$

Volume sphere = $\frac{4}{3} \pi r^3$ **volume cylinder** = $\pi r^2 h$ **volume cone** = $\pi r^2 \frac{h}{3}$

Volume cuboid = $a * b * c$ **volume pyramid** = $\frac{l * w * h}{3}$

• **Number of diagonals** = $\frac{n(n-3)}{2}$



- **Area** = $\frac{1}{2}PQ \sin(\theta) = \frac{1}{4}(b^2 + d^2 - a^2 + c^2)\tan(\theta) =$

$$\frac{1}{4}\sqrt{4P^2Q^2 - (b^2 + d^2 - a^2 - c^2)^2} =$$

$$\sqrt{(s - a)(s - b)(s - c)(s - d) - abcd\cos\frac{1}{2}(A + C)}$$

- **Area of sector** = $\frac{1}{2}r^2\theta_{rad}$

- **Area of triangle** = $\sqrt{p(p - a)(p - b)(p - c)}$, where **a, b, c** are the lengths of the three sides and $p = \frac{a+b+c}{2} = \frac{absin(C)}{2} =$

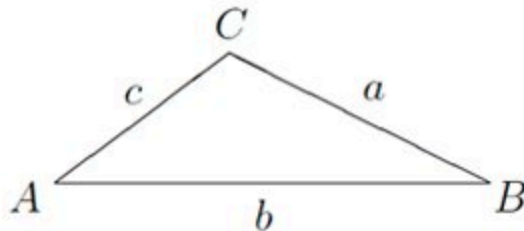
$$\left| \frac{Ax(By - Cy) + Bx(Cy - Ay) + Cx(Ay - By)}{2} \right| = \sqrt{\frac{3}{4}}S^2, \text{ where } \mathbf{s} \text{ is any}$$

$$\text{side of an equilateral triangle} = \frac{A \times B + B \times C + C \times A}{2}, \text{ where } \mathbf{A, B, C} \text{ are the three vectors}$$

- Area of an ellipse: $ab\pi$
- Lengths of a right-angled triangle given non-hypot side:

$$\frac{n^2}{4}, \frac{n^2}{4} + 1, n, n \text{ is even}, \quad \frac{n^2}{2}, \frac{n^2}{2} + 1, n, n \text{ is odd}$$

- Chord length: $2r \sin(\frac{\theta}{2})$



- $c^2 = a^2 + b^2 - 2ab \cos(C)$

- $\theta^\circ = \frac{\theta^{\text{rad}} * 180}{\pi}$

- $(1 + t)^n = \sum_{k=0}^n \binom{n}{k} t^k, t = 1 \Rightarrow 2^n$

- $(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k = \sum_{k=0}^n x^k y^{n-k}$

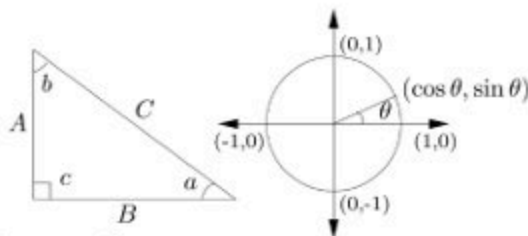
- $\left(1 + \frac{1}{n}\right)^n = 1 + \binom{n}{1} \frac{1}{n} + \binom{n}{2} \frac{1}{n^2} + \binom{n}{3} \frac{1}{n^3} + \dots + \binom{n}{n} \frac{1}{n^n}.$

- $\binom{n}{k} \frac{1}{n^k} = \frac{1}{k!} \cdot \frac{n(n-1)(n-2) \dots (n-k+1)}{n^k}$

- $\text{NPR} = \frac{n!}{(n-r)!}$

- $\text{NCR} = \frac{n!}{(n-r)! * r!}$

Trigonometry



Pythagorean theorem:

$$C^2 = A^2 + B^2.$$

Definitions:

$$\sin a = A/C, \quad \cos a = B/C,$$

$$\csc a = C/A, \quad \sec a = C/B,$$

$$\tan a = \frac{\sin a}{\cos a} = \frac{A}{B}, \quad \cot a = \frac{\cos a}{\sin a} = \frac{B}{A}.$$

Area, radius of inscribed circle:

$$\frac{1}{2}AB, \quad \frac{AB}{A+B+C}.$$

Identities:

Matrices

Multiplication:

$$C = A \cdot B, \quad c_{i,j} = \sum_{k=1}^n a_{i,k} b_{k,j}.$$

Determinants: $\det A \neq 0$ iff A is non-singular.

$$\det A \cdot B = \det A \cdot \det B,$$

$$\det A = \sum_{\pi} \prod_{i=1}^n \text{sign}(\pi) a_{i,\pi(i)}.$$

2×2 and 3×3 determinant:

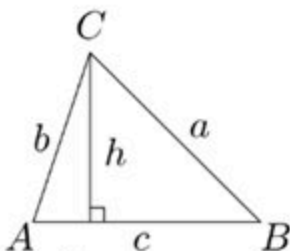
$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc,$$

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = g \begin{vmatrix} b & c \\ e & f \end{vmatrix} - h \begin{vmatrix} a & c \\ d & f \end{vmatrix} + i \begin{vmatrix} a & b \\ d & e \end{vmatrix}$$

$$= aei + bfg + cdh - ceg - fha - ibd.$$

Permanents:

$$\text{perm } A = \sum_{\pi} \prod_{i=1}^n a_{i,\pi(i)}.$$



Law of cosines:

$$c^2 = a^2 + b^2 - 2ab \cos C.$$

Area:

$$\begin{aligned} A &= \frac{1}{2}hc, \\ &= \frac{1}{2}ab \sin C, \\ &= \frac{c^2 \sin A \sin B}{2 \sin C}. \end{aligned}$$

θ	$\sin \theta$	$\cos \theta$	$\tan \theta$
0	0	1	0
$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{\pi}{2}$	1	0	∞

Important Algebraic Formulas

$$(a + b)^2 = a^2 + b^2 + 2ab$$

$$(a - b)^2 = a^2 + b^2 - 2ab$$

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^2 + b^2 = (a + b)^2 - 2ab \quad \text{or} \quad a^2 + b^2 = (a - b)^2 + 2ab$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2) = (a + b)^3 - 3ab(a + b)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2) = (a - b)^3 + 3ab(a - b)$$

$$2(a^2 + b^2) = (a + b)^2 + (a - b)^2$$

$$(a + b)^2 - (a - b)^2 = 4ab$$

$$a^4 + b^4 = (a + b)(a - b)[(a + b)^2 - 2ab]$$

$$(a - b)^2 = (a + b)^2 - 4ab$$

$$(a + b)^2 = (a - b)^2 + 4ab$$

$$a^4 + b^4 = [(a + b)^2 - 2ab]^2 - 2(ab)^2$$

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$$(a + b - c)^2 = a^2 + b^2 + c^2 + 2ab - 2bc - 2ca$$

$$(a - b - c)^2 = a^2 + b^2 + c^2 - 2ab + 2bc - 2ca$$

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$a^4 + a^2b^2 + b^4 = (a^2 + ab + b^2)(a^2 - ab + b^2)$$

$$a^4 + a^2 + 1 = (a^2 + a + 1)(a^2 - a + 1)$$

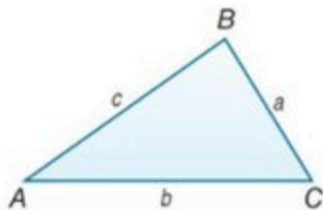
$$\text{if } a + b + c = 0 \text{ then } a^3 + b^3 + c^3 = 3abc$$

$$a^8 - b^8 = (a^4 + b^4)(a^2 + b^2)(a + b)(a - b)$$

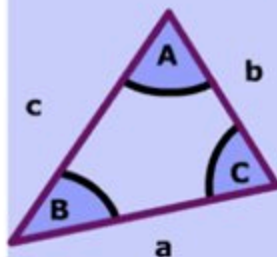
Triangle Laws

Law of Sines

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$



Law of Cosines



$$a^2 = b^2 + c^2 - 2bc \cdot \cos(A)$$

$$b^2 = a^2 + c^2 - 2ac \cdot \cos(B)$$

$$c^2 = a^2 + b^2 - 2ab \cdot \cos(C)$$

© www.mathwarehouse.com

Src: <http://www.mathwarehouse.com/trigonometry/images/law-of-cosines/law-of-cosines-formula-and-picture2.png>

$$\sin\left(\frac{\pi}{2} - \theta\right) = +\cos \theta$$

$$\cos\left(\frac{\pi}{2} - \theta\right) = +\sin \theta$$

$$\tan\left(\frac{\pi}{2} - \theta\right) = +\cot \theta$$

$$\csc\left(\frac{\pi}{2} - \theta\right) = +\sec \theta$$

$$\sec\left(\frac{\pi}{2} - \theta\right) = +\csc \theta$$

$$\cot\left(\frac{\pi}{2} - \theta\right) = +\tan \theta$$

Trigonometric formula

$$\sin (A + B) = \sin A \cos B + \sin B \cos A$$

$$\sin (A - B) = \sin A \cos B - \sin B \cos A$$

$$\cos (A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos (A - B) = \cos A \cos B + \sin A \sin B$$

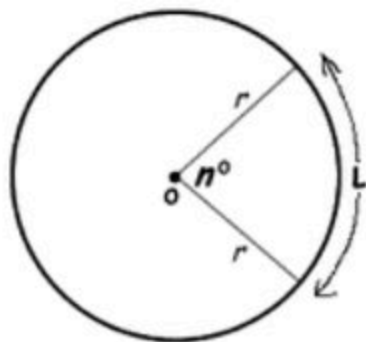
$$\tan (A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan (A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

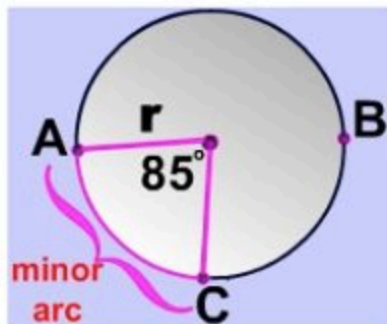
Circles

Length of an Arc Formula

$$\text{Length} = \frac{n^\circ}{360^\circ} \times 2\pi r$$



$$\text{sector area} = \frac{n}{360^\circ} \times (\pi \times r^2)$$



$\widehat{ABC}$ is the major arc
 $\widehat{AC}$ is the minor arc

I. 2D Shapes & Properties

- Rectangles

- **Perimeter:** $P = 2(l + w)$ (P : perimeter, l : length, w : width)
- **Area:** $A = lw$ (A : area, l : length, w : width)

- Squares

- **Perimeter:** $P = 4s$ (P : perimeter, s : side length)
- **Area:** $A = s^2$ (A : area, s : side length)

- Parallelograms

- **Area:** $A = bh$ (A : area, b : base, h : height)

- Rhombuses

- **Area:** $A = \frac{1}{2}d_1d_2$ (A : area, d_1, d_2 : lengths of diagonals)
- **Properties:** All four sides are equal in length.

- Triangles

- **Area:** $A = \frac{1}{2}bh$ (A : area, b : base, h : height)
- **Area (Equilateral):** $A = \frac{\sqrt{3}}{4}s^2$ (A : area, s : side length)
- **Pythagorean Theorem:** $a^2 + b^2 = c^2$ (a, b : leg lengths, c : hypotenuse length)
- **Radius of Circumcenter:** $R = \frac{abc}{4K}$ (R : circumradius, a, b, c : side lengths, K : triangle area)

- **Trapezoids**

- **Area:** $A = \frac{1}{2}(a + b)h$ (A : area, a, b : lengths of parallel bases, h : height)

- **General Polygons**

- **Perimeter:** $P = a + b + c + d...$ (P : perimeter, a, b, c, d : side lengths)
- **Sum of measures of interior angles:** $S = 180(n - 2)$ (S : sum of angles, n : number of sides)

- **Circles**

- **Circumference:** $C = 2\pi r$ or $C = \pi d$ (C : circumference, r : radius, d : diameter)
- **Area:** $A = \pi r^2$ (A : area, r : radius)
- **Circumference related to Area:** $C = \sqrt{4\pi A}$ (C : circumference, A : area)

- **Circle Segment**

- **Area:** $Area = \frac{\theta\pi r^2}{360} - \frac{1}{2}r^2 \sin \theta$ ($Area$: segment area, r : radius, θ : central angle in degrees)
- **Perimeter:** $Perimeter = 2r \sin(\theta/2) + \frac{\theta\pi r}{180}$ ($Perimeter$: segment perimeter, r : radius, θ : central angle in degrees)

II. 3D Shapes & Volumes/Surface Areas

- **Prisms**

- **Volume:** $V = Bh$ (V : volume, B : area of base, h : height)

- **Cubes**

- **Volume:** $V = s^3$ (V : volume, s : side length)

- **Cylinders**

- **Volume:** $V = \pi r^2 h$ (V : volume, r : radius of base, h : height)
- **Surface Area:** $SA = 2\pi r h + 2\pi r^2$ (SA : surface area, r : radius, h : height)

- **Spheres**

- **Volume:** $V = \frac{4}{3}\pi r^3$ (V : volume, r : radius)

- **Cones**

- **Volume:** $V = \frac{1}{3}\pi r^2 h$ (V : volume, r : radius of base, h : height)

- **Pyramids**

- **Volume:** $V = \frac{1}{3}Bh$ (V : volume, B : area of base, h : height)

Formula

If F_n is the n th Fibonacci number, then

$$F_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right)$$

Double Angle Formulas

- $\sin(2x) = 2\sin(x)\cos(x)$
- $\cos(2x) = \cos(x)^2 - \sin(x)^2 = 2\cos(x)^2 - 1 = 1 - 2\sin(x)^2$
- $\tan(2x) = \frac{2\tan(x)}{1 - \tan(x)^2}$

let two Vectors $\rightarrow v1 = (x1, y1)$, $v2 = (x2, y2)$

Cross Product

$$\text{cross} = x1 * y2 - y1 * x2$$

$\text{cross} == 0$: vectors are parallel

$\text{cross} > 0$: $v2$ is to the left of $v1$ (CCW)

$\text{cross} < 0$: $v2$ is to the right of $v1$ (CW)

Dot Product

$$\text{dot} = x1 * x2 + y1 * y2$$

$\text{dot} == 0$: vectors are perpendicular

$\text{dot} > 0$: angle between them is $< 90^\circ$

$\text{dot} < 0$: angle between them is $> 90^\circ$

If vectors are built from points:

$$AB = (Bx - Ax, By - Ay) \rightarrow (x1, y1)$$

$$AC = (Cx - Ax, Cy - Ay) \rightarrow (x2, y2)$$

$$\text{Cross} = AB.x * AC.y - AB.y * AC.x$$

$$\text{Dot} = AB.x * AC.x + AB.y * AC.y \text{ (edited)}$$

Youssef 8/5/2026 11:36 AM

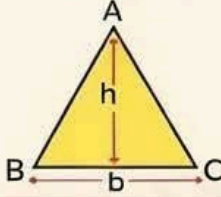
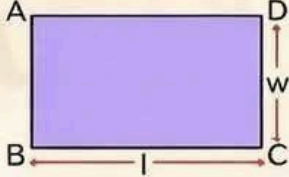
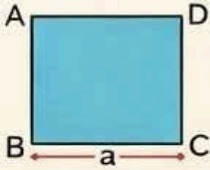
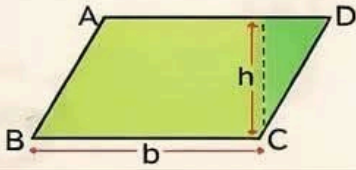
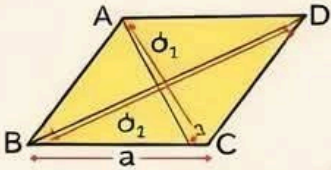
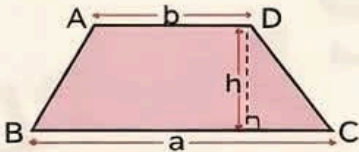
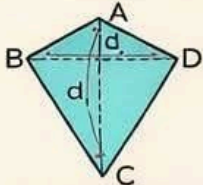
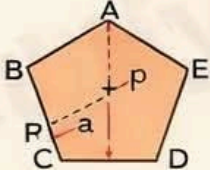
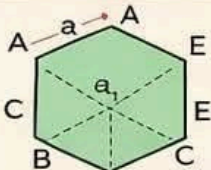
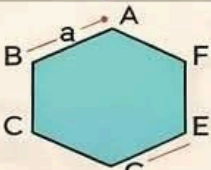
For any triangle inscribed in a circle of radius r , if the triangle has angles A, B, C (opposite to sides a, b, c respectively), then:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2r$$

```
#define PI acos(-1.0)
```

	resulting sum is divisible by 7, then the original number is divisible by 7. For example, to test divisibility of 12264 by 7, we simply check $4(1) + 6(3) + 2(2) + 2(6) + 1(4) = 4 + 18 + 4 + 12 + 4 = 42$, a two-digit number divisible by 7. Hence, 12264 must also be divisible by 7.
8	if the last three digits are evenly divisible by 8
9	if the sum is a multiple of 9 or if the sum of its digits is divisible by 9
10	if the last digit of N is 0
11	if the difference of the alternating sum of digits of N is a multiple of 11 (e.g. 2343 is divisible by 11 because $2 - 3 + 4 - 3 = 0$, which is a multiple of 11);
12	if N is divisible by both 3 and 4

Area of Polygons

Polygons	Shapes	Formulas
Triangle		$\text{Area (A)} = \frac{1}{2} (b \times h)$ here, b = base, h = height
Rectangle		$\text{Area (A)} = w \times l$ here, w = width, l = length
Square		$\text{Area (A)} = a^2$ here, a = side
Parallelogram		$\text{Area (A)} = b \times h$ here, b = base, h = height
Rhombus		$\text{Area (A)} = \frac{d_1 \times d_2}{2}$ here, d_1 and d_2 are the height diagonals
Trapezoid		$\text{Area (A)} = \frac{1}{2} (a + b) \times h$ here, a = long base b = short base, h = height
Kite		$\text{Area (A)} = \frac{d_1 \times d_2}{2}$ here, d_1 and d_2 are the diagonals
Pentagon		$\text{Area (A)} = \frac{1}{2} (p \times a)$ here, p = perimeter a = apothem
Hexagon		$\text{Area (A)} = \frac{3\sqrt{3}}{2} \times (a^2)$ here, a = side
Hexagon		$\text{Area (A)} = \frac{3\sqrt{3}}{2} \times (a^2)$ here, a = side

i	2^i	p_i
1	2	2
2	4	3
3	8	5
4	16	7
5	32	11
6	64	13
7	128	17
8	256	19
9	512	23
10	1,024	29
11	2,048	31
12	4,096	37
13	8,192	41
14	16,384	43
15	32,768	47
16	65,536	53

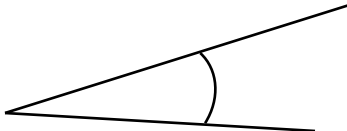
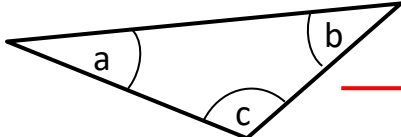
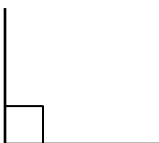
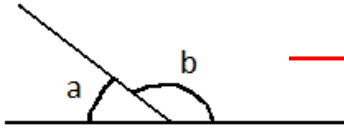
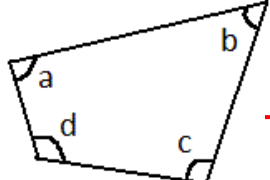
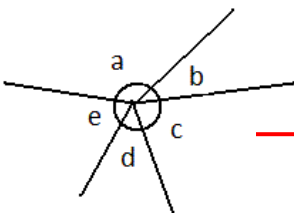
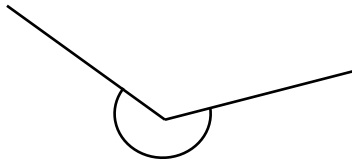
17	131,072	59
18	262,144	61
19	524,288	67
20	1,048,576	71
21	2,097,152	73
22	4,194,304	79
23	8,388,608	83
24	16,777,216	89
25	33,554,432	97
26	67,108,864	101
27	134,217,728	103
28	268,435,456	107
29	536,870,912	109
30	1,073,741,824	113
31	2,147,483,648	127
32	4,294,967,296	131

How to if a number is divisible by:

2	if the number's last digit is 0, 2, 4, 6, or 8
3	if the sum of its all digits is a multiple of 3 or divisibility by 3
4	number is divisible by 4 if its last two digits are evenly divisible by 4. For example, 2312. The last two digits are 12 and it is divisible by 4
5	if the number's last digit is either 0 or 5
6	if it is even and if the sum of its digits is divisible by 3
7	Rule 1: Remove the last digit, double it, subtract it from the truncated original number and continue doing this until only one digit remains. If this is 0 or 7, then the original number is divisible by 7. For example, to test divisibility of 12264 by 7, we simply perform the following manipulations: $1226 - 8 = 1218$ $121 - 16 = 105$ $10 - 10 = 0$ Thus, 12264 is divisible by 7. Rule 2: Take the digits of the number in reverse order, that is, from right to left, multiplying them successively by the digits 1, 3, 2, 6, 4, 5, repeating with this sequence of multipliers as long as necessary. Then add the products. If the



GEOMETRY QUICK GUIDE 1: ANGLES

Angle Types	Angle Rules																											
 Acute angle $< 90^\circ$	 Angles in a triangle add up to 180° So $a + b + c = 180^\circ$																											
 Right angle $= 90^\circ$	 Angles on a straight line add up to 180° So $a + b = 180^\circ$																											
 Obtuse angle $> 90^\circ$ and $< 180^\circ$	 Angles in a quadrilateral add up to 360° . So $a + b + c + d = 360^\circ$																											
 Straight line $= 180^\circ$	 Angles around a point add up to 360° . So $a + b + c + d + e = 360^\circ$																											
 Reflex angle $> 180^\circ$	Angles in regular shapes <table><tr><th>Name of shape</th><th>Sides</th><th>Interior angles</th></tr><tr><td>equilateral triangle</td><td>3</td><td>60°</td></tr><tr><td>square</td><td>4</td><td>90°</td></tr><tr><td>regular pentagon</td><td>5</td><td>108°</td></tr><tr><td>regular hexagon</td><td>6</td><td>120°</td></tr><tr><td>regular heptagon</td><td>7</td><td>128.6°</td></tr><tr><td>regular octagon</td><td>8</td><td>135°</td></tr><tr><td>regular nonagon</td><td>9</td><td>140°</td></tr><tr><td>regular decagon</td><td>10</td><td>144°</td></tr></table> Sum of Interior angles of regular n-sided polygons is $180(n-2)^\circ$. To find the interior angle, divide this amount by the number of sides, n.	Name of shape	Sides	Interior angles	equilateral triangle	3	60°	square	4	90°	regular pentagon	5	108°	regular hexagon	6	120°	regular heptagon	7	128.6°	regular octagon	8	135°	regular nonagon	9	140°	regular decagon	10	144°
Name of shape	Sides	Interior angles																										
equilateral triangle	3	60°																										
square	4	90°																										
regular pentagon	5	108°																										
regular hexagon	6	120°																										
regular heptagon	7	128.6°																										
regular octagon	8	135°																										
regular nonagon	9	140°																										
regular decagon	10	144°																										



Name

Date



GEOMETRY QUICK GUIDE 2: 2D SHAPES (UK)

TRIANGLES	QUADRILATERALS	REGULAR POLYGONS
Equilateral triangle All sides equal; interior angles 60°	Square All sides equal; all angles 90°	Equilateral triangle 3 sides; angle 60°
Isosceles triangle 2 sides equal; 2 congruent angles	Rectangle Opposite sides equal, all angles 90°	Square 4 sides; angle 90°
Scalene triangle No sides or angles equal	Rhombus All sides equal; 2 pairs of parallel lines; opposite angles equal	Regular Pentagon 5 sides; angle 108°
Right triangle 1 right angle	Parallelogram Opposite sides equal, 2 pairs of parallel lines	Regular Hexagon 6 sides; angle 120°
Acute triangle All angles acute	Kite Adjacent sides equal; 2 congruent angles	Regular Octagon 8 sides; angle 135°
Obtuse triangle 1 obtuse angle	Trapezium 1 pair of parallel sides	Regular Decagon 10 sides; angle 144°
	Trapezoid No pairs of parallel sides	



Free Math Sheets, Math Games and Math Help

MATH-SALAMANDERS.COM

Name

Date



GEOMETRY QUICK GUIDE 2: 2D SHAPES (UK)

TRIANGLES	QUADRILATERALS	REGULAR POLYGONS
Equilateral triangle All sides equal; interior angles 60°	Square All sides equal; all angles 90°	Equilateral triangle 3 sides; angle 60°
Isosceles triangle 2 sides equal; 2 congruent angles	Rectangle Opposite sides equal, all angles 90°	Square 4 sides; angle 90°
Scalene triangle No sides or angles equal	Rhombus All sides equal; 2 pairs of parallel lines; opposite angles equal	Regular Pentagon 5 sides; angle 108°
Right triangle 1 right angle	Parallelogram Opposite sides equal, 2 pairs of parallel lines	Regular Hexagon 6 sides; angle 120°
Acute triangle All angles acute	Kite Adjacent sides equal; 2 congruent angles	Regular Octagon 8 sides; angle 135°
Obtuse triangle 1 obtuse angle	Trapezium 1 pair of parallel sides	Regular Decagon 10 sides; angle 144°
	Trapezoid No pairs of parallel sides	

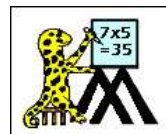


Free Math Sheets, Math Games and Math Help

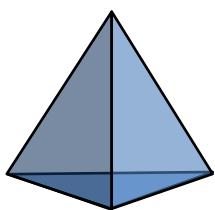
MATH-SALAMANDERS.COM

Name

Date

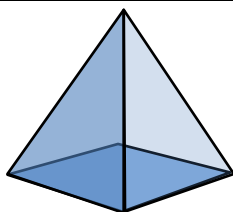


GEOMETRY QUICK GUIDE 3: 3D SHAPES



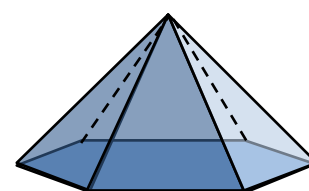
Tetrahedron

Faces: 4; Edges: 6; Vertices: 4



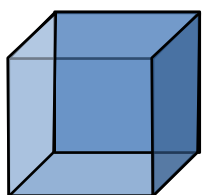
Square pyramid

Faces: 5; Edges: 8; Vertices: 5



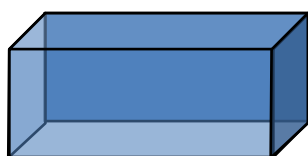
Hexagonal pyramid

Faces: 7; Edges: 12; Vertices: 7



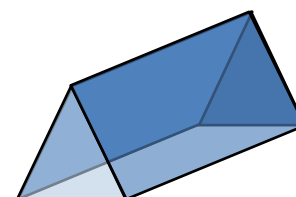
Cube

Faces: 6; Edges: 12; Vertices: 8



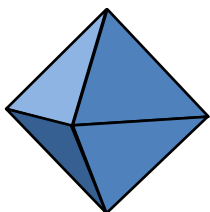
Cuboid

Faces: 6; Edges: 12; Vertices: 8



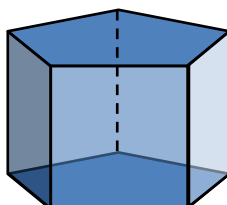
Triangular prism

Faces: 5; Edges: 9; Vertices: 6



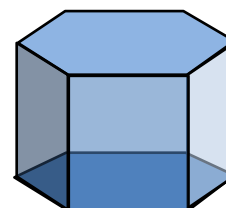
Octahedron

Faces: 8; Edges: 12; Vertices: 6



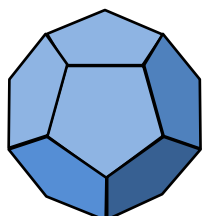
Pentagonal prism

Faces: 7; Edges: 15; Vertices: 10



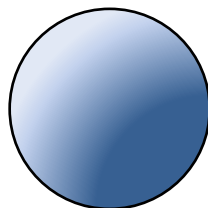
Hexagonal prism

Faces: 8; Edges: 18; Vertices: 12



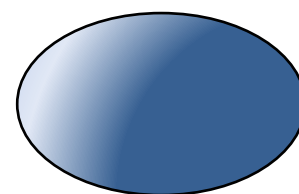
Dodecahedron

Faces: 12; Edges: 30; Vertices: 20



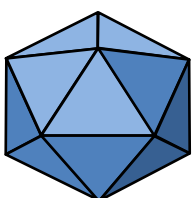
Sphere

Faces: 0 or 1; Edges: 0; Vertices: 0



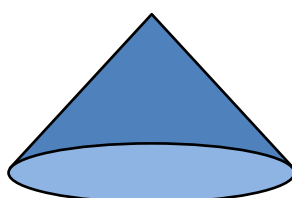
Ellipsoid

Faces: 0 or 1; Edges: 0; Vertices: 0



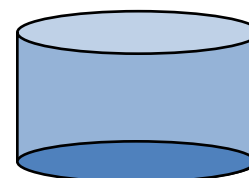
Icosahedron

Faces: 20; Edges: 30; Vertices: 12



Cone

Faces: 1 or 2; Edges: 0 or 1;
Vertices: 0 or 1



Cylinder

Faces: 2 or 3; Edges: 0 or 2;
Vertices: 0



Free Math Sheets, Math Games and Math Help

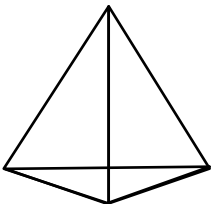
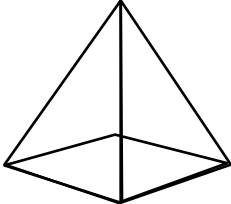
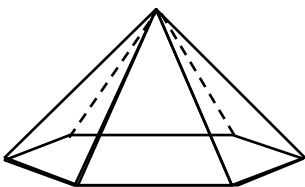
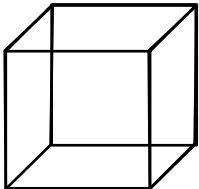
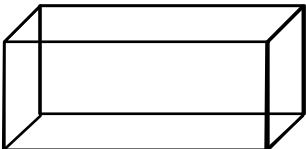
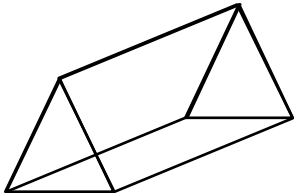
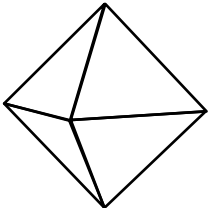
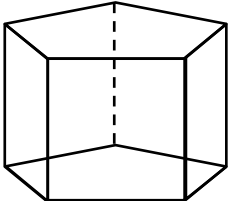
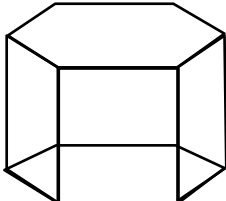
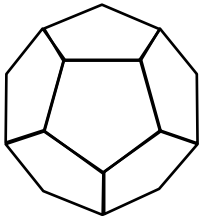
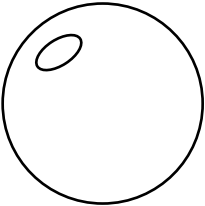
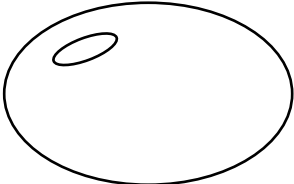
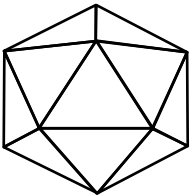
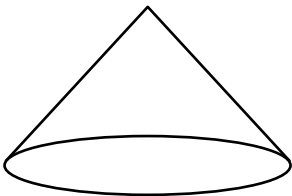
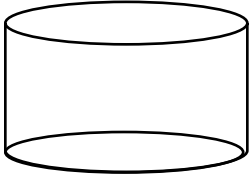
MATH-SALAMANDERS.COM

Name

Date



GEOMETRY QUICK GUIDE 3: 3D SHAPES

		
Tetrahedron Faces: 4; Edges: 6; Vertices: 4	Square pyramid Faces: 5; Edges: 8; Vertices: 5	Hexagonal pyramid Faces: 7; Edges: 12; Vertices: 7
		
Cube Faces: 6; Edges: 12; Vertices: 8	Cuboid Faces: 6; Edges: 12; Vertices: 8	Triangular prism Faces: 5; Edges: 9; Vertices: 6
		
Octahedron Faces: 8; Edges: 12; Vertices: 6	Pentagonal prism Faces: 7; Edges: 15; Vertices: 10	Hexagonal prism Faces: 8; Edges: 18; Vertices: 12
		
Dodecahedron Faces: 12; Edges: 30; Vertices: 20	Sphere Faces: 0 or 1; Edges: 0; Vertices: 0	Ellipsoid Faces: 0 or 1; Edges: 0; Vertices: 0
		
Icosahedron Faces: 20; Edges: 30; Vertices: 12	Cone Faces: 1 or 2; Edges: 0 or 1; Vertices: 0 or 1	Cylinder Faces: 2 or 3; Edges: 0 or 2; Vertices: 0



Free Math Sheets, Math Games and Math Help

MATH-SALAMANDERS.COM

Name

Date



GEOMETRY QUICK GUIDE 4: 2D SHAPES FORMULAS

TRIANGLE FORMULAS	CIRCLE FORMULAS	QUADRILATERAL FORMULAS
Angles in a triangle add up to 180° . So $a + b + c = 180^\circ$	Parts of a circle	Angles in a quadrilateral add up to 360° . So $a + b + c + d = 360^\circ$
Area of a triangle $= \frac{1}{2} \times b \times h$ or $\frac{1}{2} bh$	Circumference of a circle $= 2\pi r$ or πd	Area of a rectangle $= b \times h$ Perimeter of a rectangle $= 2b + 2h$
Pythagoras' theorem In a right triangle $a^2 + b^2 = c^2$	Area of a circle $= \pi r^2$	Area of a parallelogram $= b \times h$
Basic trigonometry laws $\sin \theta = o/h$ $\cos \theta = a/h$ $\tan \theta = o/a$	The length L of an arc $= 2\pi r(\theta/360)$ The area of a sector $= \pi r^2 (\theta/360)$	Area of a trapezium $= \frac{1}{2} (b1 + b2) \times h$



Free Math Sheets, Math Games and Math Help

MATH-SALAMANDERS.COM

Name

Date



GEOMETRY QUICK GUIDE 4: 2D SHAPES FORMULAS

TRIANGLE FORMULAS	CIRCLE FORMULAS	QUADRILATERAL FORMULAS
Angles in a triangle add up to 180° . So $a + b + c = 180^\circ$	Parts of a circle	Angles in a quadrilateral add up to 360° . So $a + b + c + d = 360^\circ$
Area of a triangle $= \frac{1}{2} \times b \times h$ or $\frac{1}{2} bh$	Circumference of a circle $= 2\pi r$ or πd	Area of a rectangle $= b \times h$ Perimeter of a rectangle $= 2b + 2h$
Pythagoras' theorem In a right triangle $a^2 + b^2 = c^2$	Area of a circle $= \pi r^2$	Area of a parallelogram $= b \times h$
Basic trigonometry laws $\sin \theta = o/h$ $\cos \theta = a/h$ $\tan \theta = o/a$	The length L of an arc $= 2\pi r(\theta/360)$ The area of a sector $= \pi r^2 (\theta/360)$	Area of a trapezium $= \frac{1}{2} (b1 + b2) \times h$



Free Math Sheets, Math Games and Math Help

MATH-SALAMANDERS.COM



GEOMETRY QUICK GUIDE 5: 3D SHAPE FORMULAS

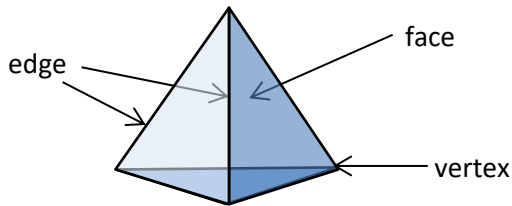
3D SHAPES

All 3d shapes can be described in terms of their faces, vertices and edges.

Face - a flat or curved surface

Edge - line where 2 faces meet

Vertex - point where 3 or more edges meet

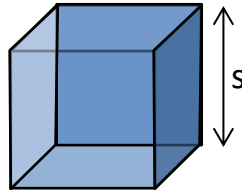


CUBE

Volume = s^3

Surface area = $6s^2$

where s is the length of one side

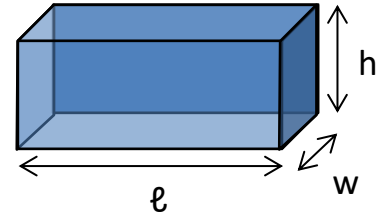


CUBOID (RECTANGULAR PRISM)

Volume = $\ell \times w \times h$

Surface area = $2\ell h + 2\ell w + 2wh$

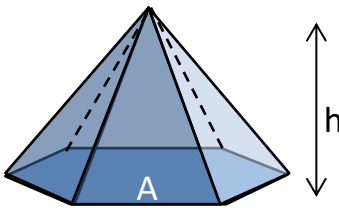
where ℓ = length, w = width, h = height



PYRAMIDS

Volume of a general pyramid = $\frac{1}{3} Ah$

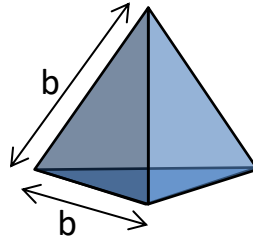
where A = base area and h = height



REGULAR TETRAHEDRON

Volume = $b^3 / 6\sqrt{2}$

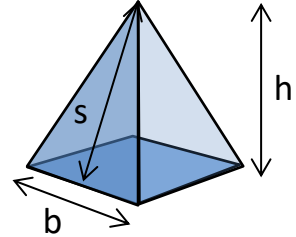
Surface area = $\sqrt{3}b^2$



RIGHT SQUARE PYRAMID

Volume = $\frac{1}{3} b^2 h$

Surface area = $b^2 + 2bs$

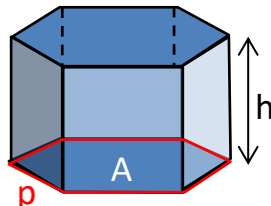
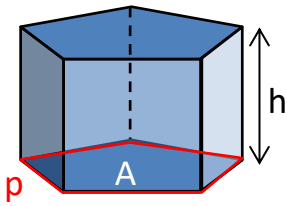


PRISMS

Volume of any prism = Ah

Surface area of a closed prism = $2A + (h \times p)$

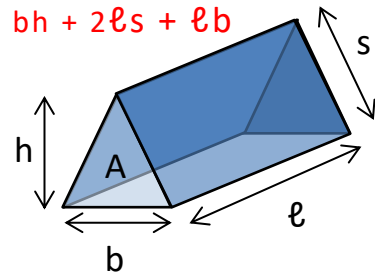
where A = base area, h = height, p = base perimeter



TRIANGULAR PRISM

Volume = $A\ell$ or $\frac{1}{2}bh\ell$

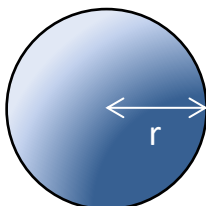
Surface area = $bh + 2\ell s + \ell b$



SPHERES

Volume = $\frac{4}{3} \pi r^3$

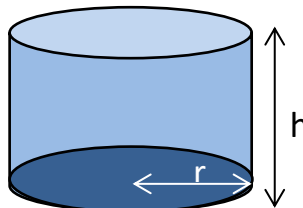
Surface area = $4\pi r^2$



RIGHT CYLINDER

Volume = $\pi r^2 h$

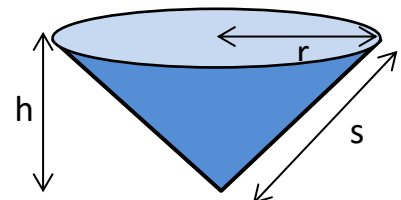
Surface area = $2\pi r(r + h)$



RIGHT CIRCULAR CONE

Volume = $\frac{1}{3} \pi r^2 h$

Surface area = $\pi r(r + s)$



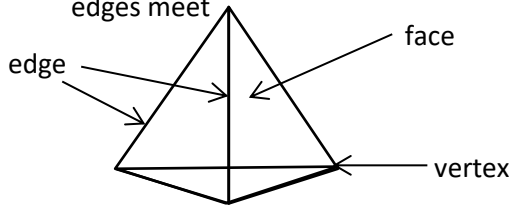


GEOMETRY QUICK GUIDE 5: 3D SHAPE FORMULAS

3D SHAPES

All 3d shapes can be described in terms of their faces, vertices and edges.

- Face - a flat or curved surface
- Edge - line where 2 faces meet
- Vertex - point where 3 or more edges meet

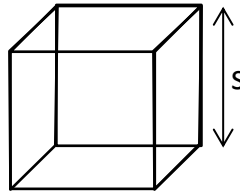


CUBE

Volume = s^3

Surface area = $6s^2$

where s is the length of one side

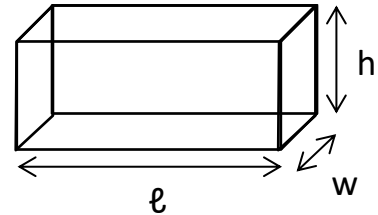


CUBOID (RECTANGULAR PRISM)

Volume = $\ell \times w \times h$

Surface area = $2\ell h + 2\ell w + 2wh$

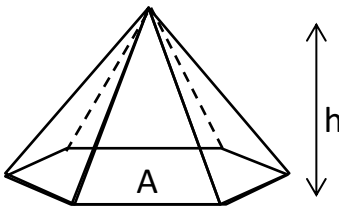
where ℓ = length, w = width, h = height



PYRAMIDS

Volume of a general pyramid = $\frac{1}{3} Ah$

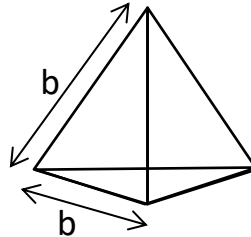
where A = base area and h = height



REGULAR TETRAHEDRON

Volume = $b^3 / 6\sqrt{2}$

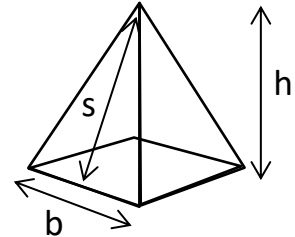
Surface area = $\sqrt{3} b^2$



RIGHT SQUARE PYRAMID

Volume = $\frac{1}{3} b^2 h$

Surface area = $b^2 + 2bs$

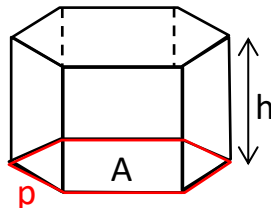
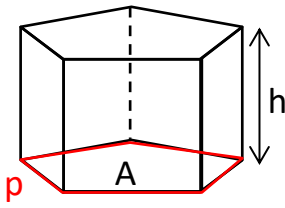


PRISMS

Volume of any prism = Ah

Surface area of a closed prism = $2A + (h \times p)$

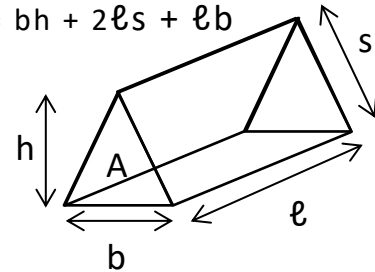
where A = base area, h = height, p = base perimeter



TRIANGULAR PRISM

Volume = $A\ell$ or $\frac{1}{2}bh\ell$

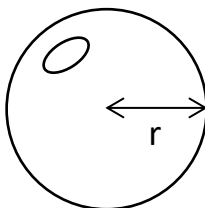
Surface area = $bh + 2\ell s + \ell b$



SPHERES

Volume = $\frac{4}{3} \pi r^3$

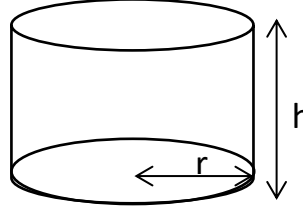
Surface area = $4\pi r^2$



RIGHT CYLINDER

Volume = $\pi r^2 h$

Surface area = $2\pi r(r + h)$



RIGHT CIRCULAR CONE

Volume = $\frac{1}{3} \pi r^2 h$

Surface area = $\pi r(r + s)$

